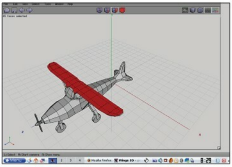

Figure 1: Top faces of wings selected

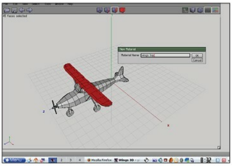

Figure 2: Naming the wings' material

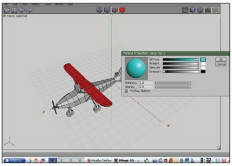

Figure 3: Assigning diffused colour to the wings' material

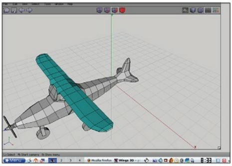

Figure 4: Top wings with separate material identity

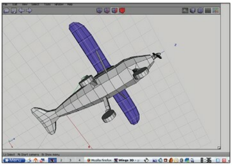

Figure 5: Lower faces of wings with separate colour

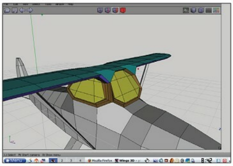

Figure 6: Assigning identity to window frames

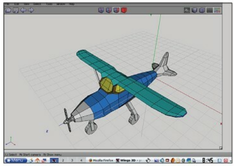

Figure 7: Body of the plane given material

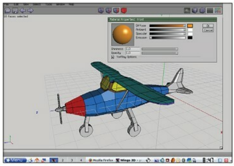

Figure 8: Nose of the plane identified

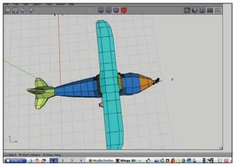

Figure 9: The tail given a material

coordinates' describes how a texture is mapped onto a 3D object.

UV texturing allows for the colouring of polygons that make up a 3D object. The resulting coloured image is called a UV texture map. The UV mapping process involves assigning pixels in the image to surface mappings on the polygon. UV is the alternative to XY; it only maps onto a texture space, rather than into the geometric space of the object. The rendering computation uses the UV texture coordinates to determine how to paint the three-dimensional surface.

When an artist creates a model using the polygonal modelling method, with a 3D modeller like Wings3D or Blender, UV coordinates are generated for each vertex in the mesh. One way is for the 3D modeller to unfold the triangle mesh at the seams, automatically laying out the triangles on a flat page. For example, if the mesh is a UV sphere, the modeller might transform it into an *equirectangular projection* [see *http://en.wikipedia.org/wiki/Equirectangular_projection*]. Once the model is unwrapped, the artist can paint a texture on each triangle individually, using the unwrapped mesh as a template.

A UV map can either be generated automatically by the 3D application, be made manually by the artist, or by some combination of both techniques. After a UV map is generated, the artist adjusts and optimises it to minimise seams and overlaps. If the model is symmetric, the artist might overlap opposite triangles to allow for painting both sides simultaneously.

The UV mapping process

The UV mapping process for the plane model involves four steps:

1. *Assigning materials to faces*: Before you start to unwrap a model, you need to assign distinct identities to various parts of the plane. Once you do this, the process of unwrapping becomes easier, since now you do not have to select the faces individually, but can use the *Outliner* (explained below) to select an entire named part on the plane.
2. *Unwrapping:* This process is done either considering the whole model, or by doing the unwrapping, part by part. The part by part method is useful in case of complex geometries—especially organic models like animals and humans.
3. *Creating the texture and saving it*: The 3D application generates a